Discrete Random Variables

Probability distributions

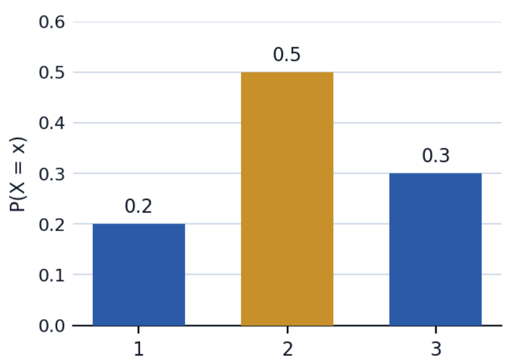
1 A discrete random variable X has the probability distribution: $P(X = 1) = 0.2$, $P(X = 2) = 0.3$, $P(X = 3) = k$, $P(X = 4) = 0.1$. Find k, using the fact that probabilities must sum to 1.

2 A discrete random variable Y has the probability distribution: $P(Y = 0) = 0.1$, $P(Y = 1) = k$, $P(Y = 2) = 2k$, $P(Y = 3) = 0.3$. Find k.

3 A discrete random variable Z has the probability distribution: $P(Z = 1) = 0.25$, $P(Z = 2) = 0.25$, $P(Z = 3) = k$, $P(Z = 4) = 0.15$, $P(Z = 5) = 0.15$. Find k.

Expected value (mean) of a discrete random variable

4 A random variable X has distribution $P(X = 1) = 0.2$, $P(X = 2) = 0.5$, $P(X = 3) = 0.3$. Find $E(X)$.



5 A random variable Y has distribution $P(Y = 0) = 0.1$, $P(Y = 1) = 0.4$, $P(Y = 2) = 0.5$. Find $E(Y)$.

6 A random variable Z has distribution $P(Z = 1) = 0.3$, $P(Z = 2) = 0.3$, $P(Z = 3) = 0.4$. Find $E(Z)$.

Variance of a discrete random variable

7 Using the distribution from the previous question ($E(X) = 2.1$), find $\text{Var}(X)$, using $\text{Var}(X) = E(X^2) - [E(X)]^2$.

8 Using the distribution $P(Y = 0) = 0.1$, $P(Y = 1) = 0.4$, $P(Y = 2) = 0.5$ ($E(Y) = 1.4$), find $\text{Var}(Y)$, using $\text{Var}(Y) = E(Y^2) - [E(Y)]^2$.

9 Using the distribution $P(Z = 1) = 0.3$, $P(Z = 2) = 0.3$, $P(Z = 3) = 0.4$ ($E(Z) = 2.1$), find $\text{Var}(Z)$, using $\text{Var}(Z) = E(Z^2) - [E(Z)]^2$.

Linear transformations of a random variable

10 A random variable X has $E(X) = 5$ and $\mathrm{Var}(X) = 3$. Find $E(2X + 4)$ and $\mathrm{Var}(2X + 4)$.

11 A random variable X has $E(X) = 8$ and $\mathrm{Var}(X) = 5$. Find $E(3X - 2)$ and $\mathrm{Var}(3X - 2)$.

12 A random variable Y has $E(Y) = 10$ and $\mathrm{Var}(Y) = 4$. Find $E(5 - 2Y)$ and $\mathrm{Var}(5 - 2Y)$.

The discrete uniform distribution

13 A fair six-sided die is rolled, and X is the number shown. Find $E(X)$ and $\mathrm{Var}(X)$, given that for a discrete uniform distribution on $1, \ldots, n$: $E(X) = \frac{n+1}{2}$ and $\mathrm{Var}(X) = \frac{n^2 - 1}{12}$.

14 A fair eight-sided die (labeled 1 to 8) is rolled, and X is the number shown. Find $E(X)$ and $\mathrm{Var}(X)$, using the same formulas.

15 A fair ten-sided die (labeled 1 to 10) is rolled, and X is the number shown. Find $E(X)$ and $\mathrm{Var}(X)$, using the same formulas.

Constructing a probability distribution from a real scenario

16 A game involves rolling two fair dice and finding the sum X. Find $P(X = 4)$ and $P(X = 7)$.

17 Using the same game of rolling two fair dice and finding the sum X, find $P(X = 9)$ and $P(X = 12)$.

18 Using the same game of rolling two fair dice and finding the sum X, find $P(X = 5)$ and $P(X = 6)$.

Cumulative distribution function

19 A random variable X has $P(X = 1) = 0.15$, $P(X = 2) = 0.35$, $P(X = 3) = 0.25$, $P(X = 4) = 0.25$. Find $P(X \leq 2)$ and $P(X > 2)$.

20 A random variable X has $P(X = 1) = 0.1$, $P(X = 2) = 0.2$, $P(X = 3) = 0.3$, $P(X = 4) = 0.4$. Find $P(X \leq 3)$ and $P(X \geq 2)$.

21 A random variable X has $P(X = 0) = 0.2$, $P(X = 1) = 0.3$, $P(X = 2) = 0.3$, $P(X = 3) = 0.2$. Find $P(X \leq 1)$ and $P(X \geq 3)$.

A well-designed word problem: a raffle ticket game

22 This question has two parts. A raffle sells 200 tickets for 5 dollars each. One ticket wins a prize of 300 dollars, and two tickets each win a prize of 50 dollars. Let X be the net profit (prize won minus the 5-dollar ticket cost) for a single ticket. (a) Write the probability distribution of X. (b) Find $E(X)$, and interpret whether buying a ticket is a fair game.